

time, floppies were the most common methods by which viruses used to get transferred as it was one of the most convenient and popular methods of backing up data and exchanging files amongst most PC users. While floppies have dipped in the popularity charts, they still are a source of worry. Especially when you are in cyber cafes and are transferring files that you have downloaded, you end up using a floppy which becomes a potential infector. Other storage media, whether CDs or CD-Rs or zip cartridges (and even hard disks when used for hard disk to hard disk transfer) become potential sources for a virus to enter your system. Just because you cannot write to a CD doesn't mean the CD cannot contain a virus (you will not be able to clean the virus from the CD, that's all).

Network Scanning

All network administrators know the FunLove virus. It took just one computer to get affected with it, and everyone in the network with some folders shared found themselves infected with the viruses or their anti-virus software popping up screens warning them about the virus.

Storage Media Safety



- Always scan floppies before you access their contents.
- Do not keep a

floppy in the floppy drive with the drive closed before booting up. If the floppy has a boot sector virus, it will automatically spread. Better still, change the BIOS setup to boot from hard disk first.

■ If you are using your floppy on some other machine to transfer data, write-protect the floppy. This will prevent your floppy from getting infected if the other system has a virus.

■ Likewise, you can also write-protect hard disks if you have taken it to your friend's place to transfer data. You can write-protect from the BIOS (which is much safer) as well as through the operating system.

■ Enable boot sector protection through the BIOS after you have finished partitioning your hard disk. This will prevent certain viruses from writing to the master boot record.



Danger Zone: Software More Prone to Attacks

Some software tools we use act as a way for viruses or malicious worms to spread. They

allow viruses to enter the system and execute without the user's permission by using an exploit. Some of these tools are:

- Microsoft Outlook (all versions prior to 2002 or 6.0)
- Microsoft Excel and Word
- Microsoft IIS 4, 5
- Sun Java Runtime Environments (old 1.0x series versions)

Mostly, any software that uses data from outside sources and has a scripting/execution part can be vulnerable to such attacks. A worm that used the Word macro feature was Melissa. Once it infected your machine, it sent an e-mail to

the first 50 addresses in your Outlook address book. This message had the subject line 'Important Message From <user-name>' and an attachment of the infected document from which the virus was launched.

Software which are more secure, or those which use a much lower degree of scripting unlinked to the core system (preferably none at all), are far less susceptible. So replacing these common utilities with more secure alternatives can prevent a possible virus attack. For e-mail, using Eudora, Netscape or other similar mailing tools that use a very limited amount of system level scripting will be a good replacement for Outlook. Disabling macro support or shifting to StarOffice can prevent Word Macro viruses from spreading.

It's a trend viruses have always followed; they move with the data. As networks become the predominant method of data transfer and floppies/other external storage mediums lose ground, many advanced viruses adopt networks as their transport medium.

In order to do so, they make use of network file shares and exploits in network services. As most networks have migrated to TCP/IP in order to connect to the Internet, it has become far easier for virus writers to write a simple code that allows the virus to hop from place to place making full use of the TCP/IP services. Most popular viruses and worms nowadays use a combination of Internet Information Server (IIS) exploits, VBScript exploits, e-mail and simple Windows file sharing to spread from one machine to another.

Keeping your software updated and having the latest exploit code patches is the only way to prevent these viruses as they operate at the service level, where an anti-virus can't detect them until the infection has actually happened.

In case of e-mail, an anti-virus mail server or a proxy would first download the mail, scan it for regular viruses as well as malicious code in the mail body, and

delete/quarantine it if any virus is found, and after that send a confirmation report to the recipient.

Most enterprise anti-virus packages also include a central console that monitors virus infections all around the network and generates alerts at a single place, from where updates can be pushed down to the individual PCs. This is also helpful in tracking down and preventing viruses that spread using the network, and prevent a full scale epidemic from happening.

As we said in the beginning, ignorance is what makes you vulnerable to viruses. Now that you have knowledge on your side, you can take the necessary precautions and go about your work with confidence. ■

MITUL MEHTA AND VARUN SINGH

